

Arabic Pro-Drop, Definiteness, and Word Order Alternation in LFG

A Computational Grammar Fragment of Modern Standard Arabic

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1. Introduction

This project implements a grammar fragment of Modern Standard Arabic (MSA) in LFG, parsed using XLE. The focus is pro-drop — the fact that Arabic verbs can stand alone as complete sentences because their morphology already encodes person, number, and gender. Alongside pro-drop, the grammar handles subject-verb agreement, VSO/SVO word order alternation via structural case, and definiteness.

One practical note upfront: XLE does not render Arabic script reliably, so the grammar and testsuite use Latin transliteration throughout. The Arabic script appears in this report and in grammar comments for transparency, but it is not what XLE processes.

2. Linguistic Phenomena and LFG Architecture

2.1 Pro-Drop and the Existential Constraint

Arabic is a pro-drop language — the subject can be omitted entirely because verb morphology encodes enough information to recover it. The verb سَمِعَ (*samia*, 'heard') already tells you the subject is third person, singular, and masculine. There is no need for an overt pronoun (Fassi Fehri 1993; Aoun, Benmamoun & Choueiri 2010).

I implemented this through an OPT-PRO template using a disjunction: either the verb inserts a null 'pro' subject directly into the f-structure, or it verifies that an overt NP has already contributed a PRED value:

```
OPT-PRO = { (^ SUBJ PRED) = 'pro'
            (^ SUBJ PRON-TYPE) = null
            (^ SUBJ CASE) = nom
            | (^ SUBJ PRED) }.
```

What I found interesting here is how little code this actually requires. Two lines in the verb entry handle both the overt and the null subject cases — the unifier does the rest. When a subject NP is present, the second disjunct fires and checks that PRED is already there. When it is absent, the first disjunct fires and builds the pro subject from scratch, inheriting gender and person from the verb form itself.

Figure 1 shows this directly: *samia al-batal al-sawt.* has an overt SUBJ with PRED 'hero', while *samia al-sawt.* has no subject NP node in the c-structure at all — yet the f-structure still has a fully specified SUBJ with PRED 'pro', PRON-TYPE null, and CASE nom.

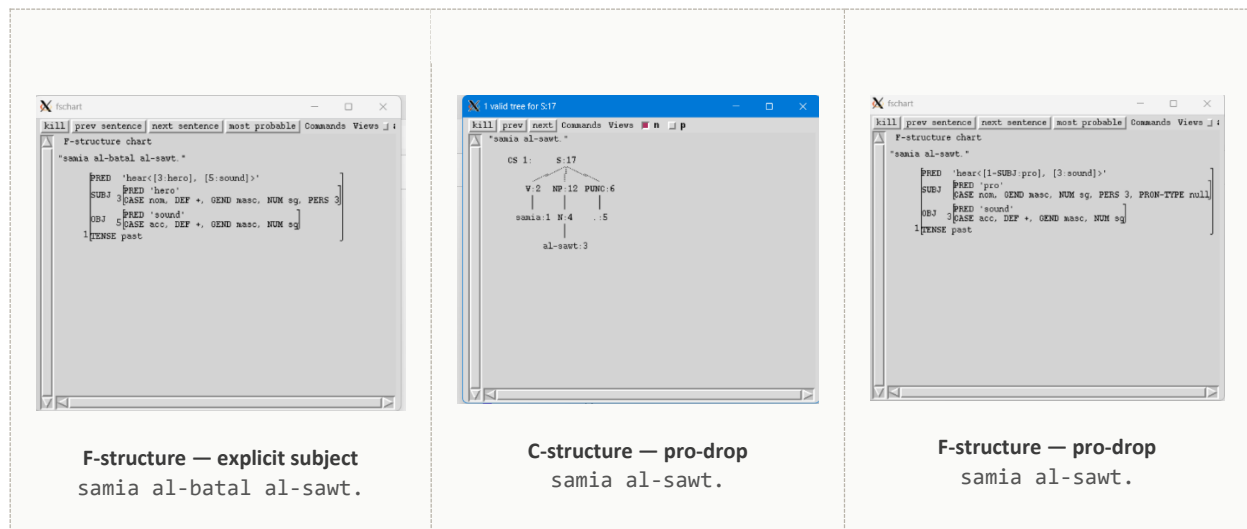


Figure 1: F-structure for the explicit-subject sentence *samia al-batal al-sawt.* (left) compared with the c-structure and f-structure for the pro-drop sentence *samia al-sawt.* (centre, right), demonstrating successful PRED 'pro' generation with automatic nominative case assignment.

2.2 Structural Case and Word Order Alternation

MSA has both VSO and SVO word order. VSO is canonical, but SVO is possible and often marks topic prominence. What allows both is case: nominative marks subjects, accusative marks objects, regardless of where they appear. I handled this with a single S rule with two disjuncts:

```
S --> { "Canonical VSO Order"
  V: ^=! ;
  (NP: (^ SUBJ) = ! (! CASE) = nom)
  (NP: (^ OBJ) = ! (! CASE) = acc)
| "Marked SVO Order"
  NP: (^ SUBJ) = ! (! CASE) = nom ;
  V: ^=! ;
  (NP: (^ OBJ) = ! (! CASE) = acc)
} (PUNC) .
```

The key point is that the case equations are identical in both disjuncts — so it does not matter whether the verb is first or second. The unifier assigns SUBJ and OBJ based on case, not position. This is deliberately simpler than a full morphological implementation: rather than inflecting every noun for case, I use structural assignment and keep the lexicon manageable.

Figure 2 shows *al-batal samia al-sawt.* — the SVO version. The c-structure has NP before V, but the f-structure is identical to the VSO parse: *al-batal* is SUBJ, *al-sawt* is OBJ. The grammar does not care which came first.

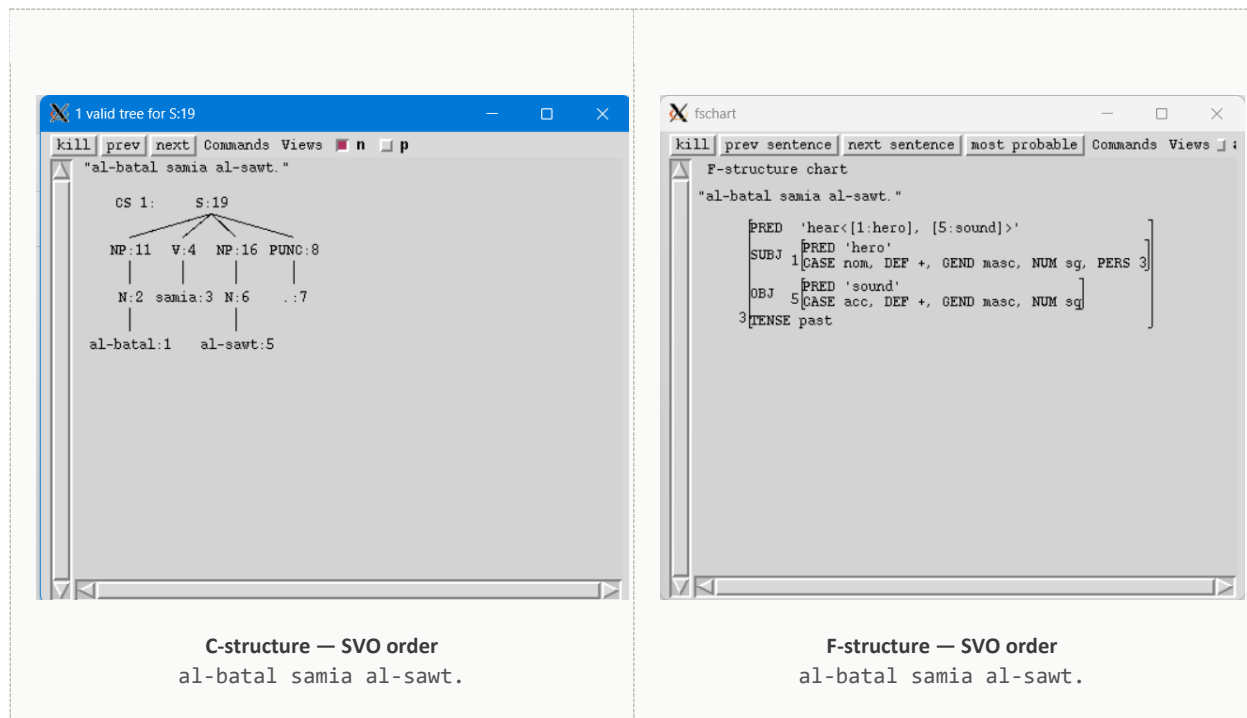


Figure 2: C-structure and f-structure of the marked SVO sentence *al-batal samia al-sawt.*, demonstrating that *al-batal* is correctly assigned SUBJ and *al-sawt* is correctly assigned OBJ via structural case rather than linear position.

2.3 Definiteness and Morphological Scope

Arabic marks definiteness with the prefix *al-*. I did not implement a morphological analyser for this — that would have gone well beyond the project's scope. Instead, I entered definite and indefinite noun forms as separate lexical entries, each assigned ($\wedge$ DEF) = + or ($\wedge$ DEF) = - via a lightweight template. It is a simplification, but it keeps the analysis clean and focused on the syntax.

3. Grammar Coverage and Design Decisions

The grammar covers five verbs — *samia/samiat* 'heard' (transitive), *faqada/faqadat* 'lost' (transitive), *taraka/tarakat* 'left' (transitive), *haraba/harabat* 'fled' (intransitive), *saqata/saqatat* 'fell' (intransitive) — and six nouns across both genders: *al-batal*, *al-rajul*, *al-thill*, *al-sawt* (masculine), *al-rooh*, *al-ghurfah* (feminine). I chose a gothic-themed lexicon deliberately — 'the shadow fled,' 'the spirit left the room' — partly because it makes the agreement violations feel less arbitrary, and partly because it is more interesting to work with than the usual 'the student wrote the letter.'

Each test sentence targets exactly one dimension: valency (transitive vs. intransitive), gender agreement (masculine vs. feminine verb form), or subject realisation (overt vs. null). This made debugging much more straightforward — when a sentence failed unexpectedly, I knew immediately which constraint to look at. The five negative sentences each violate exactly one constraint, so they function as minimal-pair tests of the grammar's restrictions.

One limitation worth being upfront about: the grammar does not model full vocalisation or the VSO agreement asymmetry — in MSA, a verb preceding a plural subject takes singular morphology, which

would require additional machinery. These are real Arabic facts I had to leave out to keep the scope reasonable.

4. Evaluation

I ran the full testsuite using `parse-testfile`. The results were exactly what I expected: 15 good sentences with exactly one parse each, and 5 sentences with 0 parses. Figure 3 shows the full output.

```

english@Ahmad: /mnt/c/User  X  +  v
% parse-testfile arabic_prodrop.testsuite.lfg
parsing sentences 1 to end in arabic_prodrop.testsuite.lfg...
((1) (1 0.006 10) (3 words))
((2) (1 0.003 10) (2 words))
((3) (1 0.003 7) (2 words))
((4) (1 0.003 4) (1 word))
((5) (1 0.003 10) (3 words))
((6) (1 0.002 10) (2 words))
((7) (1 0.003 10) (3 words))
((8) (1 0.002 10) (2 words))
((9) (1 0.003 7) (2 words))
((10) (1 0.002 4) (1 word))
((11) (1 0.005 10) (3 words))
((12) (1 0.003 10) (3 words))
((13) (1 0.009 7) (2 words))
((14) (1 0.002 10) (3 words))
((15) (1 0.003 10) (3 words))
((16) (0 0.003 0) (3 words))
((17) (0 0.003 0) (4 words))
((18) (0 0.004 5) (3 words))
((19) (0 0.002 5) (2 words))
((20) (0 0.002 5) (3 words))
done parsing arabic_prodrop.testsuite.lfg
XLE release of Mar 17, 2017 21:12.
Grammar = /mnt/c/Users/engli/desktop/uni/GDss/finalpro/arabic_prodrop.lfg.
Grammar last modified on Aug 15, 2026 16:41.
Host machine is Ahmad.
20 sentences, 0 errors, 0 mismatches
5 sentences had 0 parses
15 good sentences
15 sentences had exactly one grammatical parse
timeout = 100
max_xle_scratch_storage = 100 MB
prune_subtree_location = 1
max_new_events_per_graph_when_skimming = 500
maximum raw subtrees per sentence = 15 (#1)
maximum event count per sentence = 393
average event count per graph = 39.215
tokens = 4.8%, morph = 0.0%, multiword = 0.0%, lex = 0.0%, chart = 17.8%,
unifier = 77.3%, completer = 0.0%, solver = 0.0%, output = 0.0%
max_alloc_mem = 52.02 MB
0.066 CPU secs total, 0.009 CPU secs max
elapsed time = 2 seconds
the stats of the following exclude failed sentences:

```

Figure 3: XLE `parse-testfile` output for the full twenty-sentence testsuite, confirming 0 errors and the expected pass/fail pattern across all positive and negative examples.

5. Conclusion

The grammar shows that pro-drop in Arabic can be handled cleanly in LFG with a two-line disjunction in the verb's lexical entry — no empty categories in the rules needed. What surprised me is how well the structural case approach handles word order alternation: the same f-structure comes out whether the verb is first or the subject is. The main thing I would extend if I continued this project is plural agreement, where Arabic gets considerably more complex and the VSO/SVO asymmetry becomes much harder to ignore.

6. References

- Aoun, J., Benmamoun, E. & Choueiri, L. (2010). *The Syntax of Arabic*. Cambridge University Press.
- Butt, M., King, T. H., Niño, M.-E. & Segond, F. (1999). *A Grammar Writer's Cookbook*. CSLI Publications.
- Dalrymple, M. (2001). *Lexical Functional Grammar*. Academic Press.
- Fassi Fehri, A. (1993). *Issues in the Structure of Arabic Clauses and Words*. Kluwer Academic Publishers.

Appendix: Testsuite and Leipzig Glosses

Abbreviations: 1/2/3 = first/second/third person; SG = singular; MASC = masculine; FEM = feminine; PST = past tense; DEF = definite; NOM = nominative; ACC = accusative.

A.1 Grammatical — VSO, Explicit Subjects vs. Pro-Drop

(1) سَمِعَ البَطْلُ الصَّوْتَ.

samia *al-batal* *al-sawt.*
hear.PST.3SG.MASC DEF-hero.NOM DEF-sound.ACC
'The hero heard the sound.'

(2) سَمِعَ الصَّوْتَ.

samia *al-sawt.*
hear.PST.3SG.MASC DEF-sound.ACC
'He heard the sound.'

(3) هَرَبَ الظِّل.

haraba *al-thill.*
flee.PST.3SG.MASC DEF-shadow.NOM
'The shadow fled.'

(4) هَرَبَ.

haraba.
flee.PST.3SG.MASC
'He/it fled.'

(5) فَقَدَ الرَّجُلُ الظِّل.

faqada *al-rajul* *al-thill.*
lose.PST.3SG.MASC DEF-man.NOM DEF-shadow.ACC
'The man lost the shadow.'

(6) فَقَدَ الظِّل.

faqada *al-thill.*
lose.PST.3SG.MASC DEF-shadow.ACC
'He lost the shadow.'

A.2 Grammatical — VSO, Feminine Agreement

(7) تَرَكَّتِ الرُّوحُ العُرْفَةَ.

tarakat *al-rooh* *al-ghurfah.*
leave.PST.3SG.FEM DEF-spirit.NOM DEF-room.ACC
'The spirit left the room.'

(8) تَرَكَّتِ الْغُرْفَةَ.

tarakat al-ghurfah.
 leave.PST.3SG.FEM DEF-room.ACC
 'She/it left the room.'

(9) سَقَطَتِ الرُّوحُ.

saqatat al-rooh.
 fall.PST.3SG.FEM DEF-spirit.NOM
 'The spirit fell.'

(10) سَقَطَتْ.

saqatat.
 fall.PST.3SG.FEM
 'She/it fell.'

A.3 Grammatical — SVO Alternation

(11) الْبَطْلُ سَمِعَ الصَّوْتَ.

al-batal samia al-sawt.
 DEF-hero.NOM hear.PST.3SG.MASC DEF-sound.ACC
 'The hero heard the sound.'

(12) الرُّوحُ تَرَكَّتِ الْغُرْفَةَ.

al-rooh tarakat al-ghurfah.
 DEF-spirit.NOM leave.PST.3SG.FEM DEF-room.ACC
 'The spirit left the room.'

(13) الظِّلُّ هَرَبَ.

al-thill haraba.
 DEF-shadow.NOM flee.PST.3SG.MASC
 'The shadow fled.'

A.4 Grammatical — Definiteness Contrast

(14) سَمِعَ بَطْلٌ الصَّوْتَ.

samia batal al-sawt.
 hear.PST.3SG.MASC hero.INDEF.NOM DEF-sound.ACC
 'A hero heard the sound.'

(15) سَمِعَ الْبَطْلُ صَوْتًا.

samia al-batal sawt.
 hear.PST.3SG.MASC DEF-hero.NOM sound.INDEF.ACC
 'The hero heard a sound.'

A.5 Ungrammatical — Valency Violations

(16) *هَرَبَ البَطْلُ الصَّوْتِ.

* *haraba* *al-batal* *al-sawt*.
 flee.PST.3SG.MASC DEF-hero.NOM DEF-sound.ACC
intended: 'The hero fled the sound.'

Ungrammatical: haraba is intransitive and cannot subcategorise for an object.

(17) *سَمِعَ البَطْلُ الصَّوْتِ العُرْفَةَ.

* *samia* *al-batal* *al-sawt* *al-ghurfah*.
 hear.PST.3SG.MASC DEF-hero.NOM DEF-sound.ACC DEF-room.ACC
intended: 'The hero heard the sound the room.'

Ungrammatical: samia is transitive and cannot license a second object.

A.6 Ungrammatical — Agreement Violations

(18) *سَمِعَتِ البَطْلُ الصَّوْتِ.

* *samiat* *al-batal* *al-sawt*.
 hear.PST.3SG.FEM DEF-hero.NOM DEF-sound.ACC
intended: 'The hero heard the sound.'

Ungrammatical: feminine verb agreement with a masculine subject.

(19) *سَقَطَ الرُّوحِ.

* *saqata* *al-rooh*.
 fall.PST.3SG.MASC DEF-spirit.NOM
intended: 'The spirit fell.'

Ungrammatical: masculine verb agreement with a feminine subject.

(20) *تَرَكَ الرُّوحِ العُرْفَةَ.

* *taraka* *al-rooh* *al-ghurfah*.
 leave.PST.3SG.MASC DEF-spirit.NOM DEF-room.ACC
intended: 'The spirit left the room.'

Ungrammatical: masculine verb agreement with a feminine subject.